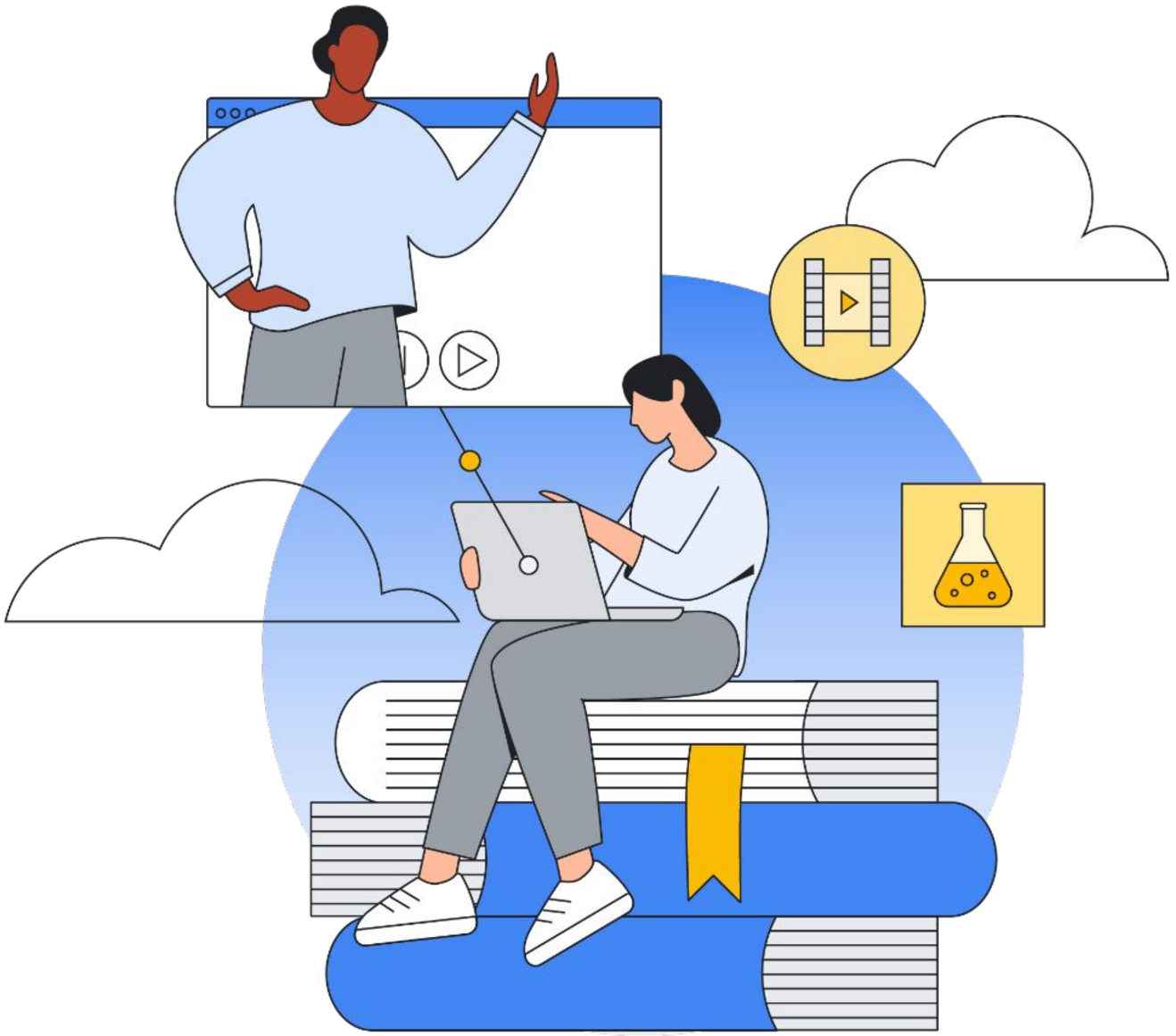
Google Cloud

Skill Badge Session

Deploy a RAG application with vector search in Firestore



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Thank you!

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What will you do in this challenge lab?

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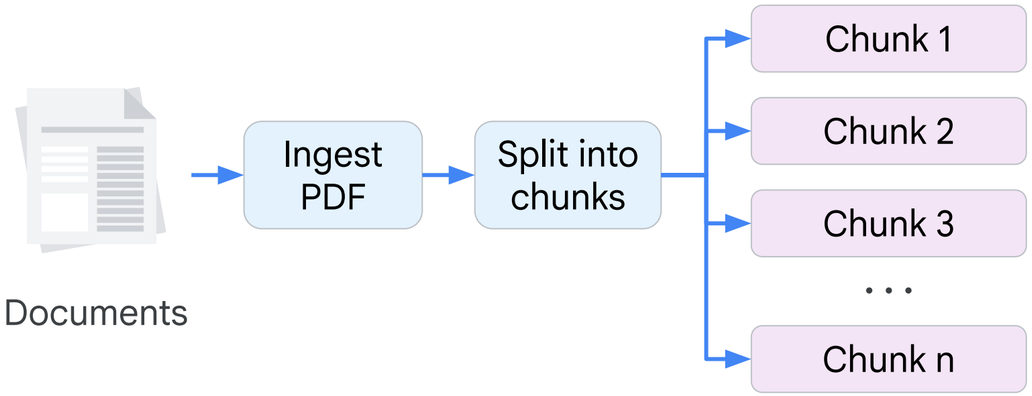
Load a text document and split it into chunks



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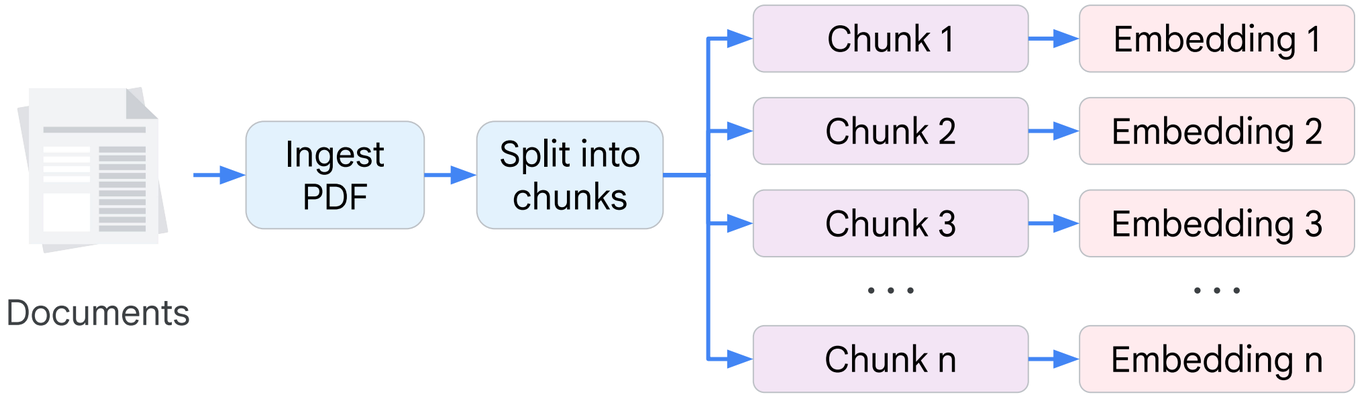
Generate embeddings for each chunk



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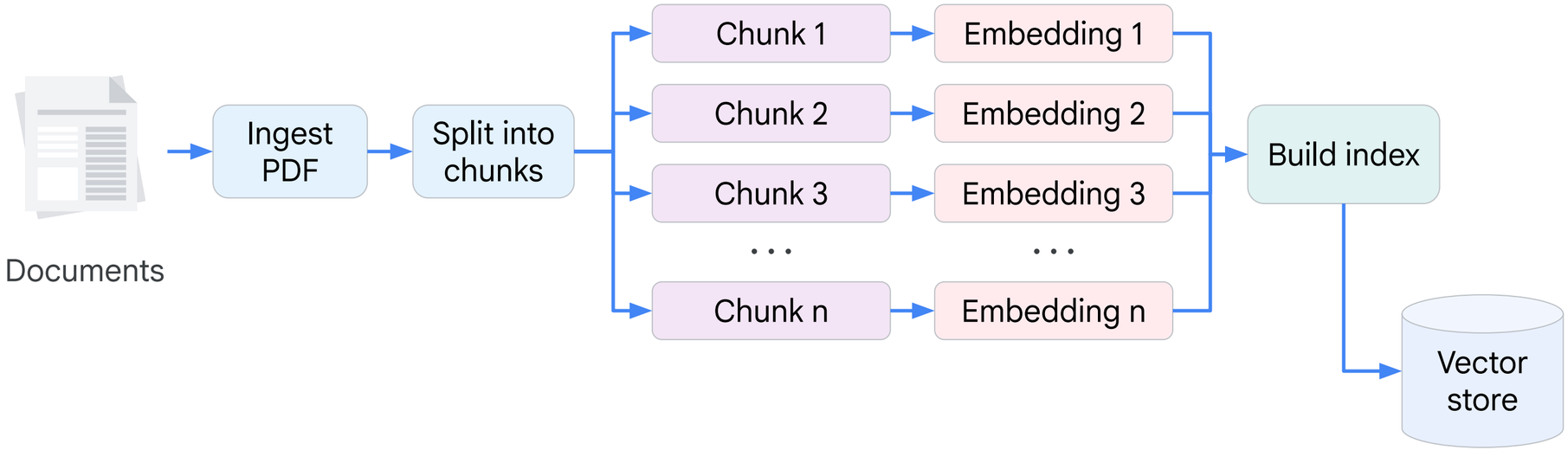
Store the text chunks and their embeddings



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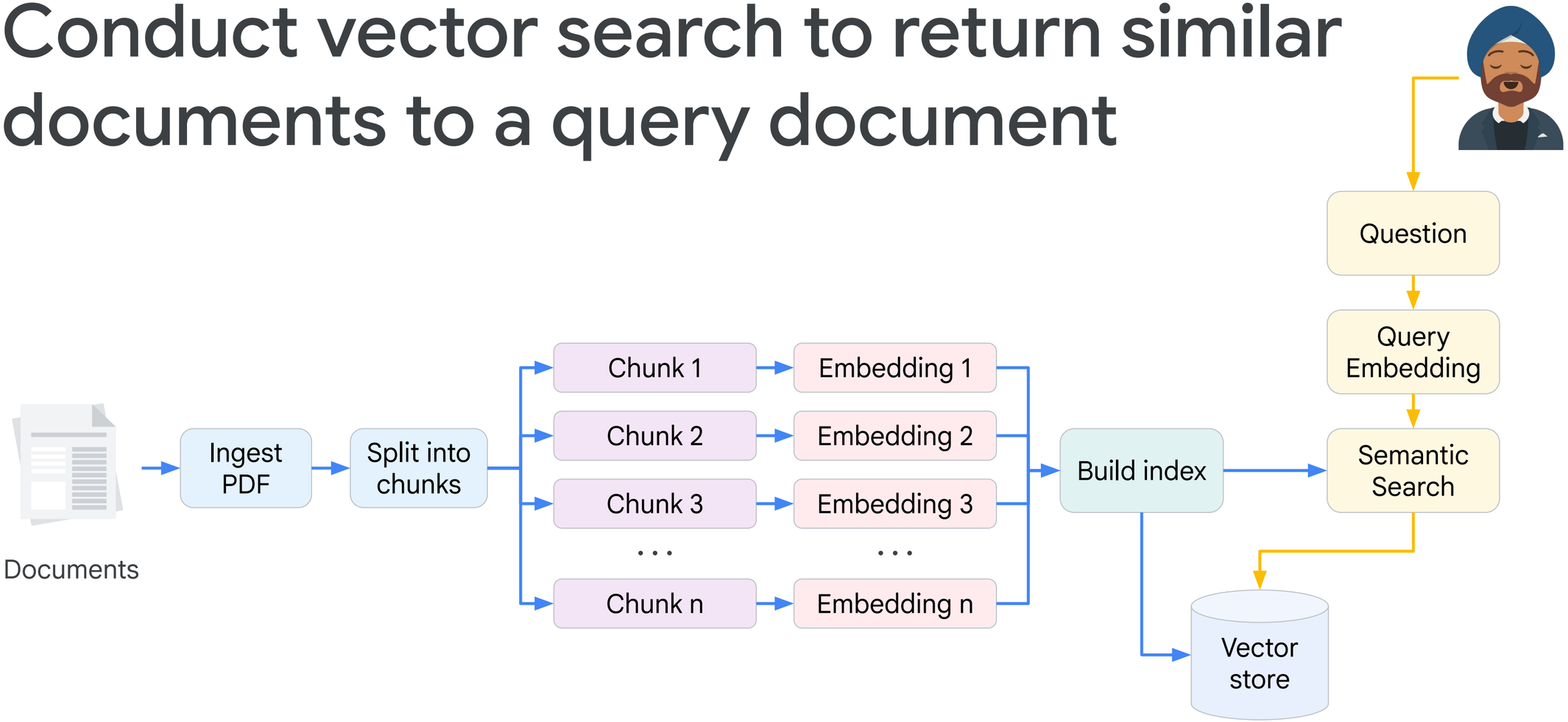
Conduct vector search to return similar documents to a query document



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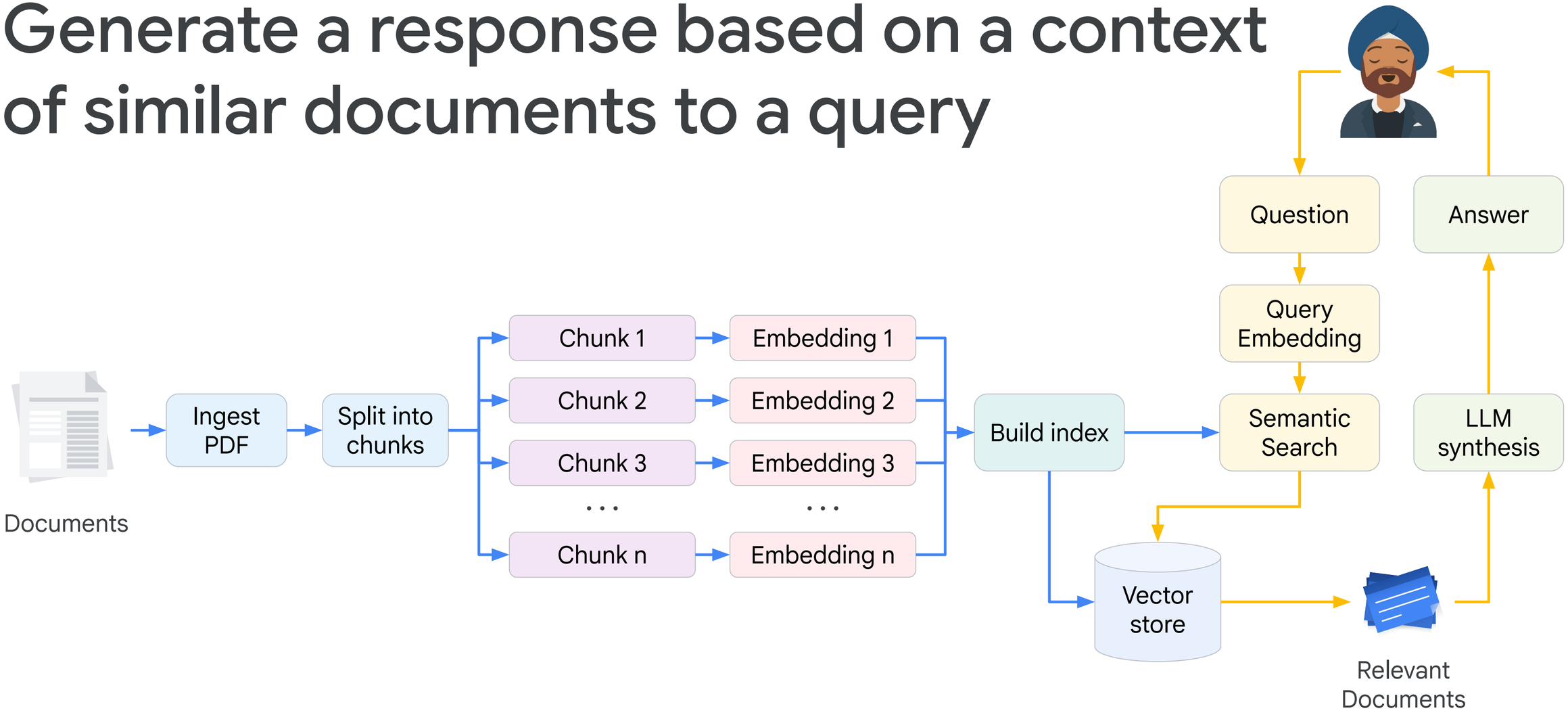
Generate a response based on a context of similar documents to a query



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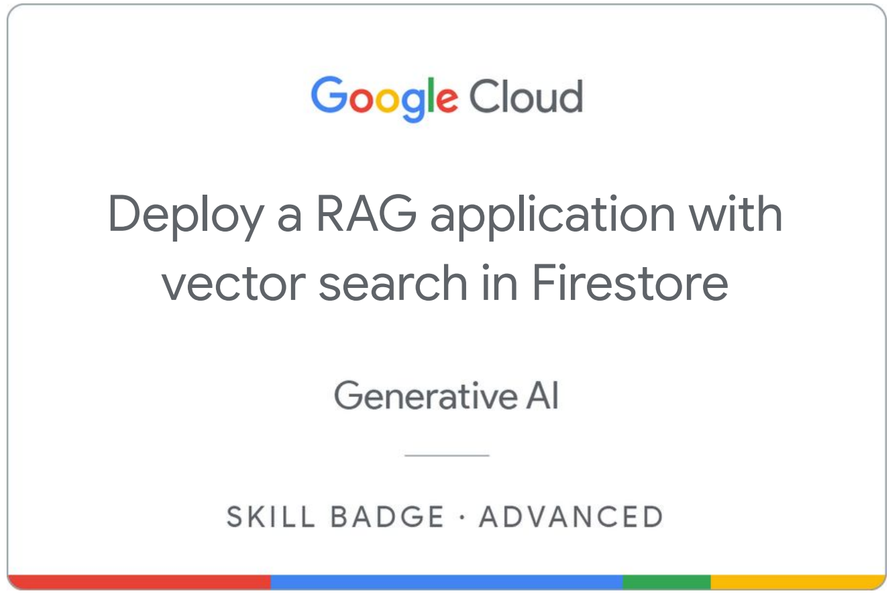

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From the Sign In screen, use the
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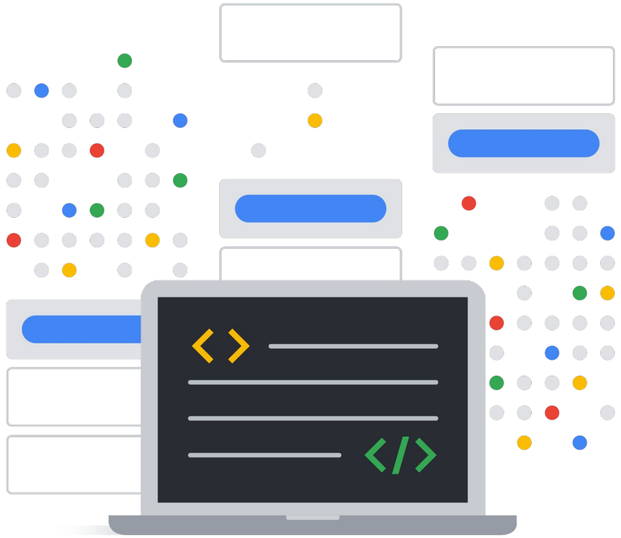

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This skill badge is part of the learning path: Build with Vertex AI: Building RAG Applications with Vertex AI

1. **Course**: Create Embeddings, Vector Search, and RAG with BigQuery
2. **Course**: Implement RAG with Vertex AI
3. **Course**: Implement Hybrid Search
4. **Challenge Lab**: Deploy a RAG application with vector search in Firestore

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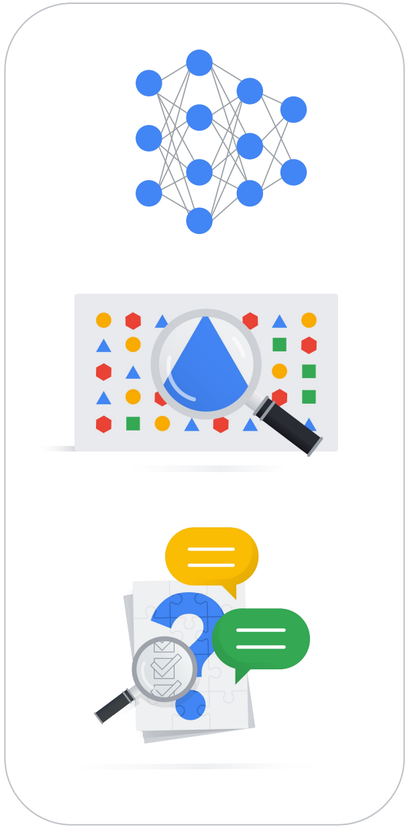
Create Embeddings, Vector Search, and RAG with BigQuery

Course



- Retrieval Augmented Generation (RAG) in BigQuery.
- Create embeddings, search a vector space, and generate improved answers.
- Build a RAG pipeline using BigQuery, GenAI models and embedding models.

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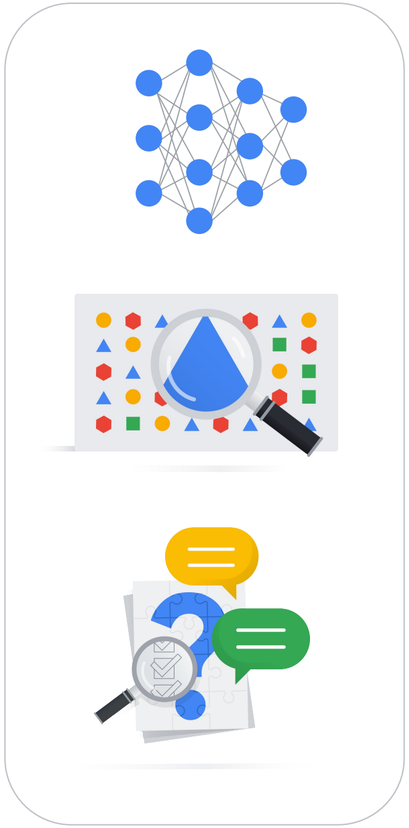
Implement RAG with Vertex AI

Course



- The practicalities of building a RAG solution on Vertex AI.
- How LangChain provides a workflow for RAG components.
- Chunking, embedding and vectorizing data, creating retrievers and agents, and running queries.
- Explore ways to enhance RAG performance.

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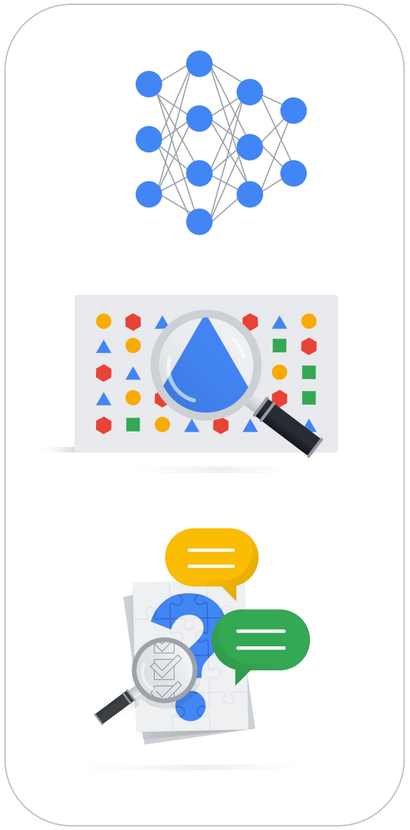
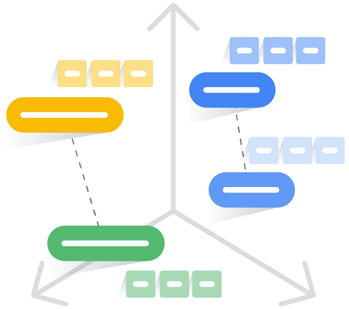
Implement Hybrid Search

Course





- Explain the purpose of Hybrid Search.
- Differentiate between sparse (keyword) and dense (semantic) embeddings for searching.
- Run hybrid searches.
- Analyze the benefits of combining semantic and keyword search techniques.

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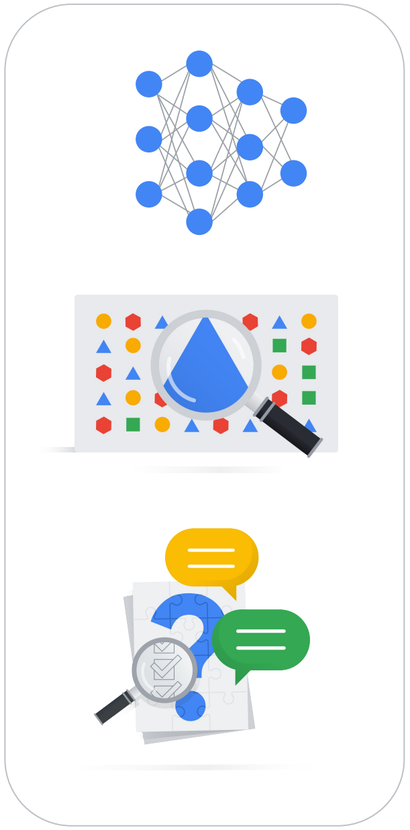
Deploy a RAG application with vector search in Firestore

Challenge Lab



- Load a text document and split it into chunks.
- Generate embeddings for each chunk.
- Store the text chunks and their embeddings.
- Conduct vector search to return similar documents to a query document.
- Generate a response based on a context of similar documents to a query.

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You can search for this course on the Explore tab of Cloud Skills Boost for Partners.

Learning Path:
Build with Vertex AI: Building RAG Applications with Vertex AI

Skill Badge:
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